QuizPilot AI Study & Performance Report

- Session Title: CS-402 Distributed Systems & Security Synthetic Practice Run
- Session ID: sess_sample_synthetic_2026
- Report ID: rep_sess_sample_synthetic_2026
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A. Executive Summary

Metric	Value
Total Questions Processed	11
Answered / Unanswered	10 answered / 1 unanswered
Verified Correct	8
Verified Incorrect	1
AI-Suggested (Unverified)	1
Uncertain / Ambiguous	1
Unsupported Layout	0
Verified Performance Score	88.9% (8/9 verified)

> Scoring Integrity Note: Calculated strictly over 9 officially verified question(s) (8/9). Excluded 2 unverified/AI-suggested/uncertain question(s) from the score denominator.

Topic Coverage
- Consensus & Replication
- Web & Browser Security
- Transactions & Indexing
- HTTP & Transport Protocols
- Complexity & Graph Algorithms
- Fault Tolerance & Idempotency
- Authentication & Cryptography

Principal Strengths
- Consensus & Replication (100% verified accuracy)
- Web & Browser Security (100% verified accuracy)
- Transactions & Indexing (100% verified accuracy)
- Complexity & Graph Algorithms (100% verified accuracy)

Principal Weaknesses & Knowledge Gaps
- HTTP & Transport Protocols (1 verified error(s), 50% accuracy)
- Fault Tolerance & Idempotency (1 unverified/uncertain, 0 unanswered)
- Authentication & Cryptography (1 unverified/uncertain, 0 unanswered)

B. Complete Question Bank

Q1. [q_std_1] According to the CAP theorem, when a distributed data store experiences a network partition (P), which trade-off is unavoidable?
- Type: single_choice | Subject: Distributed Systems | Topic: Consensus & Replication (Raft, Paxos & Quorum Consistency)
- Difficulty: MEDIUM (5/10) | Importance Priority: LOW
- Verification Status: verified_correct | Source: official_feedback (Official In-Quiz Practice Feedback) | Confidence: 99%
- Options:
  - A (opt_cap_a): Choosing between linearizable Consistency (C) and 100% Availability (A)
  - B (opt_cap_b): Choosing between Durability and Horizontal Sharding
  - C (opt_cap_c): Guaranteeing both strict linearizability and zero-latency writes on all nodes
  - D (opt_cap_d): Disabling TLS encryption to preserve TCP throughput
- Selected Response: opt_cap_a
- Official Correct Answer: opt_cap_a
- Explanation: During a network partition (P), nodes that cannot communicate must either reject/delay requests to preserve linearizable Consistency (CP) or serve potentially stale data to remain Available (AP).
- Priority Rationale: Low Priority: Verified correct response on an isolated or already mastered detail in "Consensus & Replication" with no recurring errors.
- Recommended Revision Action: Schedule lightweight 14-day maintenance review only.

Q2. [q_std_2] In Chrome Extension Manifest V3, which component replaces persistent background pages for event-driven lifecycle coordination?
- Type: single_choice | Subject: Cybersecurity | Topic: Web & Browser Security (CSP, XSS, CSRF & Same-Origin Policy)
- Difficulty: EASY (5/10) | Importance Priority: LOW
- Verification Status: verified_correct | Source: official_feedback (Official In-Quiz Practice Feedback) | Confidence: 99%
- Options:
  - A (opt_mv3_a): DOM Frameset Controller
  - B (opt_mv3_b): Background Service Worker
  - C (opt_mv3_c): ActiveX NPAPI Plugin
  - D (opt_mv3_d): Unrestricted Synchronous XMLHttpRequest Worker
- Selected Response: opt_mv3_b
- Official Correct Answer: opt_mv3_b
- Explanation: Manifest V3 replaces persistent background pages with event-driven Service Workers to improve security and memory efficiency.
- Priority Rationale: Low Priority: Verified correct response on an isolated or already mastered detail in "Web & Browser Security" with no recurring errors.
- Recommended Revision Action: Schedule lightweight 14-day maintenance review only.

Q3. [q_std_3] Which database recovery mechanism guarantees Atomicity and Durability by flushing redo/undo log records to disk before dirty pages are written?
- Type: single_choice | Subject: Database Systems | Topic: Transactions & Indexing (ACID, Isolation Levels & B-Tree/WAL)
- Difficulty: MEDIUM (5/10) | Importance Priority: LOW
- Verification Status: verified_correct | Source: official_feedback (Official In-Quiz Practice Feedback) | Confidence: 99%
- Options:
  - A (opt_wal_a): Write-Ahead Logging (WAL)
  - B (opt_wal_b): Read Uncommitted Dirty Cache
  - C (opt_wal_c): UDP Broadcast Gossip
  - D (opt_wal_d): Client-Side LocalStorage Polling
- Selected Response: opt_wal_a
- Official Correct Answer: opt_wal_a
- Explanation: Write-Ahead Logging (WAL) ensures that all modifications are recorded in stable log storage before data pages are updated in-place.
- Priority Rationale: Low Priority: Verified correct response on an isolated or already mastered detail in "Transactions & Indexing" with no recurring errors.
- Recommended Revision Action: Schedule lightweight 14-day maintenance review only.

Q4. [q_std_4] Which of the following HTTP methods are defined as idempotent by RFC 9110? (Select all that apply)
- Type: multiple_choice | Subject: Networking & Systems | Topic: HTTP & Transport Protocols (TCP/UDP, HTTP/2, HTTP/3 & DNS)
- Difficulty: HARD (5.8/10) | Importance Priority: HIGH
- Verification Status: verified_incorrect | Source: official_feedback (Official In-Quiz Practice Feedback) | Confidence: 99%
- Options:
  - A (opt_http_get): GET
  - B (opt_http_put): PUT
  - C (opt_http_post): POST
  - D (opt_http_delete): DELETE
- Selected Response: opt_http_get, opt_http_put
- Official Correct Answer: opt_http_get, opt_http_put, opt_http_delete
- Explanation: Per RFC 9110 Section 9.2.2, GET, HEAD, OPTIONS, TRACE, PUT, and DELETE are idempotent; POST is not idempotent.
- Priority Rationale: High Priority: Verified error on a hard-difficulty foundational concept in "HTTP & Transport Protocols".
- Recommended Revision Action: Compare your selected response (opt_http_get, opt_http_put) against the verified key (opt_http_get, opt_http_put, opt_http_delete) and restudy "TCP/UDP, HTTP/2, HTTP/3 & DNS".

Q5. [q_std_5] Which defense-in-depth controls directly mitigate Cross-Site Scripting (XSS) when rendering untrusted page text in a browser dashboard? (Select all that apply)
- Type: multiple_choice | Subject: Cybersecurity | Topic: Web & Browser Security (CSP,

XSS, CSRF & Same-Origin Policy)
- Difficulty: MEDIUM (6.8/10) | Importance Priority: LOW
- Verification Status: verified_correct | Source: official_feedback (Official In-Quiz Practice Feedback) | Confidence: 99%
- Options:
  - A (opt_xss_a): Context-aware HTML output encoding and safe DOM APIs (textContent)
  - B (opt_xss_b): Strict Content Security Policy (CSP) disallowing unsafe-inline scripts
  - C (opt_xss_c): Passing raw extracted HTML directly into innerHTML and eval()
  - D (opt_xss_d): Sanitizing and stripping executable script/iframe tags prior to storage
- Selected Response: opt_xss_a, opt_xss_b, opt_xss_d
- Official Correct Answer: opt_xss_a, opt_xss_b, opt_xss_d
- Explanation: Safe DOM sinks (textContent), strict CSP headers, and input sanitization prevent untrusted data from executing as code.
- Priority Rationale: Low Priority: Verified correct response on an isolated or already mastered detail in "Web & Browser Security" with no recurring errors.
- Recommended Revision Action: Schedule lightweight 14-day maintenance review only.

Q6. [q_std_6] True or False: Binary search over a sorted array of n elements has a worst-case time complexity of O(log n).
- Type: true_false | Subject: Algorithms & Data Structures | Topic: Complexity & Graph Algorithms (Asymptotic Analysis, Trees & Shortest Paths)
- Difficulty: EASY (4/10) | Importance Priority: LOW
- Verification Status: verified_correct | Source: official_feedback (Official In-Quiz Practice Feedback) | Confidence: 99%
- Options:
  - A (opt_tf_true): True
  - B (opt_tf_false): False
- Selected Response: opt_tf_true
- Official Correct Answer: opt_tf_true
- Explanation: Binary search halves the remaining search space on each comparison, completing in at most floor(log2(n)) + 1 comparisons.
- Priority Rationale: Low Priority: Verified correct response on an isolated or already mastered detail in "Complexity & Graph Algorithms" with no recurring errors.
- Recommended Revision Action: Schedule lightweight 14-day maintenance review only.

Q7. [q_std_7] True or False: In Raft consensus, a follower node can unilaterally commit log entries without a majority quorum if its local clock is synchronized via NTP.
- Type: true_false | Subject: Distributed Systems | Topic: Consensus & Replication (Raft, Paxos & Quorum Consistency)
- Difficulty: MEDIUM (6.3/10) | Importance Priority: LOW
- Verification Status: verified_correct | Source: official_feedback (Official In-Quiz Practice Feedback) | Confidence: 99%
- Options:
  - A (opt_raft_true): True

  - B (opt_raft_false): False
- Selected Response: opt_raft_false
- Official Correct Answer: opt_raft_false
- Explanation: Raft safety requires a leader elected by a majority quorum and replication across a majority of cluster nodes before committing any log entry.
- Priority Rationale: Low Priority: Verified correct response on an isolated or already mastered detail in "Consensus & Replication" with no recurring errors.
- Recommended Revision Action: Schedule lightweight 14-day maintenance review only.

Q8. [q_std_8] Enter the numeric HTTP status code returned when a client exceeds the permitted request rate limit (Too Many Requests):
- Type: text_input | Subject: Networking & Systems | Topic: HTTP & Transport Protocols (TCP/UDP, HTTP/2, HTTP/3 & DNS)
- Difficulty: EASY (5.5/10) | Importance Priority: LOW
- Verification Status: verified_correct | Source: official_feedback (Official In-Quiz Practice Feedback) | Confidence: 99%
- Selected Response: 429
- Official Correct Answer: 429
- Explanation: RFC 6585 defines HTTP status code 429 as Too Many Requests.
- Priority Rationale: Low Priority: Verified correct response on an isolated or already mastered detail in "HTTP & Transport Protocols" with no recurring errors.
- Recommended Revision Action: Schedule lightweight 14-day maintenance review only.

Q9. [q_std_9] In Raft leader election, what minimum fraction of cluster votes must a candidate receive to become leader? (Type "majority")
- Type: text_input | Subject: Distributed Systems | Topic: Consensus & Replication (Raft, Paxos & Quorum Consistency)
- Difficulty: MEDIUM (6.5/10) | Importance Priority: LOW
- Verification Status: verified_correct | Source: official_feedback (Official In-Quiz Practice Feedback) | Confidence: 99%
- Selected Response: majority
- Official Correct Answer: majority
- Explanation: A Raft candidate must receive votes from a strict majority ((N/2) + 1) of nodes in the cluster.
- Priority Rationale: Low Priority: Verified correct response on an isolated or already mastered detail in "Consensus & Replication" with no recurring errors.
- Recommended Revision Action: Schedule lightweight 14-day maintenance review only.

Q10. [q_std_10] When designing retry logic for transient RPC timeouts in distributed microservices, which strategy prevents synchronized retry storms across clients?
- Type: single_choice | Subject: Distributed Systems | Topic: Fault Tolerance & Idempotency (Retries, Backoff & Circuit Breakers)
- Difficulty: MEDIUM (5/10) | Importance Priority: MEDIUM
- Verification Status: ai_suggested | Source: instructor_note (CS-402 Instructor Lecture Notes Fault Tolerance Week 4) | Confidence: 85%
- Options:

  - A (opt_retry_a): Exponential backoff combined with randomized jitter and bounded retry limits
  - B (opt_retry_b): Unbounded zero-delay tight-loop retries on every client thread
  - C (opt_retry_c): Fixed 100ms synchronized cron retries across all worker nodes
- Selected Response: opt_retry_a
- Official Correct Answer: _Not verified by an official key_
- Explanation: Instructor Explanation: Exponential backoff with randomized jitter spreads out client retries and prevents cascading thundering-herd overload. (Note: Sourced from instructor lecture notes; not yet verified by an official grading key.)
- Priority Rationale: Medium Priority: Supporting concept in "Fault Tolerance & Idempotency" with ai suggested status requiring authoritative verification.
- Recommended Revision Action: Validate the AI-proposed candidate answer against an official answer key before relying on it for exam preparation.

Q11. [q_std_11] Which experimental quantum-resistant lattice parameter set is optimal for custom sub-millisecond IoT mesh handshakes without hardware acceleration?
- Type: single_choice | Subject: Cybersecurity | Topic: Authentication & Cryptography (OAuth2, JWT, TLS & Key Management)
- Difficulty: HARD (5/10) | Importance Priority: MEDIUM
- Verification Status: uncertain | Source: approved_bank (Community Study Bank (Conflicting Errata #11)) | Confidence: 25%
- Options:
  - A (opt_q_a): Parameter Set Alpha-512
  - B (opt_q_b): Parameter Set Beta-768
  - C (opt_q_c): Parameter Set Gamma-1024
- Selected Response: _None (Unanswered)_
- Official Correct Answer: _Not verified by an official key_
- Explanation: Conflicting reference key: Study Guide v1 specifies Alpha-512 whereas Errata Sheet v2 specifies Beta-768.
- Priority Rationale: Medium Priority: Supporting concept in "Authentication & Cryptography" with uncertain status requiring authoritative verification.
- Recommended Revision Action: Obtain an authoritative explanation or answer key to resolve this ambiguous/unanswered item.

C. Important-Question Classification

> Methodology: Questions are classified into High, Medium, and Low priority using transparent criteria: (1) Verified errors or repeated concepts across supplied materials = High; (2) Unverified AI suggestions, uncertain keys, or unanswered items = Medium; (3) Isolated details already verified correct = Low.
>
> Disclaimer: Priority ratings reflect conceptual centrality and observed session gaps. QuizPilot AI does not claim that any specific question will appear on a future examination without validated historical frequency evidence.

- High Priority: 1 | Medium Priority: 2 | Low Priority: 8

- [HIGH] q_std_4 (HTTP & Transport Protocols hard): High Priority: Verified error on a hard-difficulty foundational concept in "HTTP & Transport Protocols". -> *Action:* Compare your selected response (opt_http_get, opt_http_put) against the verified key (opt_http_get, opt_http_put, opt_http_delete) and restudy "TCP/UDP, HTTP/2, HTTP/3 & DNS".
- [MEDIUM] q_std_10 (Fault Tolerance & Idempotency medium): Medium Priority: Supporting concept in "Fault Tolerance & Idempotency" with ai suggested status requiring authoritative verification. -> *Action:* Validate the AI-proposed candidate answer against an official answer key before relying on it for exam preparation.
- [MEDIUM] q_std_11 (Authentication & Cryptography hard): Medium Priority: Supporting concept in "Authentication & Cryptography" with uncertain status requiring authoritative verification. -> *Action:* Obtain an authoritative explanation or answer key to resolve this ambiguous/unanswered item.
- [LOW] q_std_1 (Consensus & Replication medium): Low Priority: Verified correct response on an isolated or already mastered detail in "Consensus & Replication" with no recurring errors. -> *Action:* Schedule lightweight 14-day maintenance review only.
- [LOW] q_std_2 (Web & Browser Security easy): Low Priority: Verified correct response on an isolated or already mastered detail in "Web & Browser Security" with no recurring errors. -> *Action:* Schedule lightweight 14-day maintenance review only.
- [LOW] q_std_3 (Transactions & Indexing medium): Low Priority: Verified correct response on an isolated or already mastered detail in "Transactions & Indexing" with no recurring errors. -> *Action:* Schedule lightweight 14-day maintenance review only.
- [LOW] q_std_5 (Web & Browser Security medium): Low Priority: Verified correct response on an isolated or already mastered detail in "Web & Browser Security" with no recurring errors. -> *Action:* Schedule lightweight 14-day maintenance review only.
- [LOW] q_std_6 (Complexity & Graph Algorithms easy): Low Priority: Verified correct response on an isolated or already mastered detail in "Complexity & Graph Algorithms" with no recurring errors. -> *Action:* Schedule lightweight 14-day maintenance review only.
- [LOW] q_std_7 (Consensus & Replication medium): Low Priority: Verified correct response on an isolated or already mastered detail in "Consensus & Replication" with no recurring errors. -> *Action:* Schedule lightweight 14-day maintenance review only.
- [LOW] q_std_8 (HTTP & Transport Protocols easy): Low Priority: Verified correct response on an isolated or already mastered detail in "HTTP & Transport Protocols" with no recurring errors. -> *Action:* Schedule lightweight 14-day maintenance review only.
- [LOW] q_std_9 (Consensus & Replication medium): Low Priority: Verified correct response on an isolated or already mastered detail in "Consensus & Replication" with no recurring errors. -> *Action:* Schedule lightweight 14-day maintenance review only.

D. Performance Analysis

- Verified Accuracy: 88.9% (8/9)
- Excluded from Score: 2 question(s) (Questions with verificationStatus in {ai_suggested, uncertain, unanswered, unsupported} without an authoritative answer key are excluded from the verified accuracy denominator to prevent unverified guesses from skewing the score.)
- Average Time per Question: 5 ms (Total: 55 ms)
- Confidence Calibration: Detected 1 high-confidence incorrect response(s) (overconfidence alert). Prioritize reviewing those misconceptions.

Accuracy by Topic

Subject	Topic	Total	Verified	Correct	Incorrect	Uncertain/AI	Verified Accuracy	Avg Time (ms)
Distributed Systems	Consensus & Replication	3	3	3	0	0	100%	5
Cybersecurity	Web & Browser Security	2	2	2	0	0	100%	5
Database Systems	Transactions & Indexing	1	1	1	0	0	100%	5
Networking & Systems	HTTP & Transport Protocols	2	2	1	1	0	50%	5
Algorithms & Data Structures	Complexity & Graph Algorithms	1	1	1	0	0	100%	5
Distributed Systems	Fault Tolerance & Idempotency	1	0	0	0	1	Unverified	5
Cybersecurity	Authentication & Cryptography	1	0	0	0	1	Unverified	5

Error Rate by Question Type

Question Type	Total	Verified Total	Verified Incorrect	Verified Error Rate	Uncertain/AI
single_choice	5	3	0	0%	2
multiple_choice	2	2	1	50%	0
true_false	2	2	0	0%	0
text_input	2	2	0	0%	0

E. Personalized Preparation Plan

- Learner Daily Study Budget: 75 minutes/day
- Target Assessment Date: 2026-11-09 (14 days remaining)
- Recommended Revision Order: HTTP & Transport Protocols Fault Tolerance & Idempotency Authentication & Cryptography Consensus & Replication Web & Browser Security Transactions & Indexing Complexity & Graph Algorithms

Priority Topics for Revision

1. HTTP & Transport Protocols (25 min/day) Prioritized due to 1 verified error(s).
2. Fault Tolerance & Idempotency (25 min/day) Prioritized due to 1 unverified/uncertain item(s).

3. Authentication & Cryptography (25 min/day) Prioritized due to 1 unverified/uncertain item(s).
4. Consensus & Replication (25 min/day) Prioritized due to 3 foundational question(s) reviewed.
5. Web & Browser Security (25 min/day) Prioritized due to 2 foundational question(s) reviewed.
6. Transactions & Indexing (25 min/day) Prioritized due to 1 foundational question(s) reviewed.
7. Complexity & Graph Algorithms (25 min/day) Prioritized due to 1 foundational question(s) reviewed.

Suggested Daily Study Schedule
- Day 1 (2026-10-26) [75 min] Focus: HTTP & Transport Protocols, Fault Tolerance & Idempotency
  - Review official explanations & error logs for HTTP & Transport Protocols (30 min)
  - Complete targeted active-recall drill on HTTP & Transport Protocols & Fault Tolerance & Idempotency (30 min)
  - Resolve uncertain/unverified flashcards in Spaced Repetition Queue (15 min)
- Day 2 (2026-10-27) [75 min] Focus: Fault Tolerance & Idempotency, Authentication & Cryptography
  - Review official explanations & error logs for Fault Tolerance & Idempotency (30 min)
  - Complete targeted active-recall drill on Fault Tolerance & Idempotency & Authentication & Cryptography (30 min)
  - Resolve uncertain/unverified flashcards in Spaced Repetition Queue (15 min)
- Day 3 (2026-10-28) [75 min] Focus: Authentication & Cryptography, Consensus & Replication
  - Review official explanations & error logs for Authentication & Cryptography (30 min)
  - Complete targeted active-recall drill on Authentication & Cryptography & Consensus & Replication (30 min)
  - Resolve uncertain/unverified flashcards in Spaced Repetition Queue (15 min)
- Day 4 (2026-10-29) [75 min] Focus: Consensus & Replication, Web & Browser Security
  - Review official explanations & error logs for Consensus & Replication (30 min)
  - Complete targeted active-recall drill on Consensus & Replication & Web & Browser Security (30 min)
  - Resolve uncertain/unverified flashcards in Spaced Repetition Queue (15 min)
- Day 5 (2026-10-30) [75 min] Focus: Web & Browser Security, Transactions & Indexing
  - Review official explanations & error logs for Web & Browser Security (30 min)
  - Complete targeted active-recall drill on Web & Browser Security & Transactions & Indexing (30 min)
  - Resolve uncertain/unverified flashcards in Spaced Repetition Queue (15 min)
- Day 6 (2026-10-31) [75 min] Focus: Transactions & Indexing, Complexity & Graph Algorithms
  - Review official explanations & error logs for Transactions & Indexing (30 min)
  - Complete targeted active-recall drill on Transactions & Indexing & Complexity &

Graph Algorithms (30 min)
  - Resolve uncertain/unverified flashcards in Spaced Repetition Queue (15 min)
- Day 7 (2026-11-01) [75 min] Focus: Complexity & Graph Algorithms, HTTP & Transport Protocols
    - Review official explanations & error logs for Complexity & Graph Algorithms (30 min)
  - Complete targeted active-recall drill on Complexity & Graph Algorithms & HTTP & Transport Protocols (30 min)
  - Resolve uncertain/unverified flashcards in Spaced Repetition Queue (15 min)

Spaced-Repetition Recommendations
- HTTP & Transport Protocols (immediate_24h, intervals: Day 1, Day 3, Day 7, Day 14, Day 30): Contains 1 verified incorrect response(s); requires immediate 24-hour re-testing before extending intervals. (Questions: q_std_4, q_std_8)
- Fault Tolerance & Idempotency (short_term_3d, intervals: Day 2, Day 5, Day 10, Day 21): Contains 1 unverified or unanswered question(s); verify against official key within 48-72 hours. (Questions: q_std_10)
- Authentication & Cryptography (short_term_3d, intervals: Day 2, Day 5, Day 10, Day 21): Contains 1 unverified or unanswered question(s); verify against official key within 48-72 hours. (Questions: q_std_11)
- Consensus & Replication (consolidation_14d, intervals: Day 7, Day 14, Day 30): All verified items answered accurately; schedule maintenance recall at 7 and 14 days. (Questions: q_std_1, q_std_7, q_std_9)
- Web & Browser Security (consolidation_14d, intervals: Day 7, Day 14, Day 30): All verified items answered accurately; schedule maintenance recall at 7 and 14 days. (Questions: q_std_2, q_std_5)
- Transactions & Indexing (consolidation_14d, intervals: Day 7, Day 14, Day 30): All verified items answered accurately; schedule maintenance recall at 7 and 14 days. (Questions: q_std_3)
- Complexity & Graph Algorithms (consolidation_14d, intervals: Day 7, Day 14, Day 30): All verified items answered accurately; schedule maintenance recall at 7 and 14 days. (Questions: q_std_6)

Follow-Up Mock-Test Plan
- Timing: After completing Day 3 revision blocks (3 days from today)
- Focus Areas: HTTP & Transport Protocols, Fault Tolerance & Idempotency, Authentication & Cryptography
- Target Conditions: Timed practice session with official answer key loaded, targeting >= 90% verified accuracy and 0 unresolved high-priority gaps.

Criteria for Measuring Improvement
- [ ] Achieve >= 90% accuracy on questions backed by an official answer key.
- [ ] Reduce High-Confidence Incorrect responses (overconfidence errors) to 0.
- [ ] Convert 100% of Uncertain / AI-Suggested items in High-Priority topics to Verified status using authoritative references.
- [ ] Maintain average response time under 45 seconds per question without skipping

multi-select option checks.